Polyatomic Ions Quiz

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Section 1: Identification (Match the Following)

1. Match the following polyatomic ions to their correct formulas:

   - a) Nitrate
   - b) Carbonate
   - c) Ammonium
   - d) Sulfate
   - e) Hydroxide

   Options:

   - i) NO_3^-
   - ii) CO_3^{2-}
   - iii) NH_4^+
   - iv) SO_4^{2-}
   - v) OH^-

2. Match the following names to their formulas:

   - a) Phosphate
   - b) Sulfite
   - c) Chromate
   - d) Permanganate
   - e) Cyanide

Options:

- i) PO_4^{3-}
- ii) SO_3^{2-}
- iii) CrO_4^{2-}
- iv) MnO_4^-
- v) CN^-

Section 2: True/False

3. The formula for the hypochlorite ion is ClO^-.

4. The dichromate ion has the formula $Cr_2O_7^{2-}$.

5. Sulfate and sulfite have the same charge.

6. The ammonium ion is a cation with a $+1$ charge.

7. The acetate ion can be written as CH_3COO^- or $C_2H_3O_2^-$.

8. Hydroxide ion is neutral.

9. Phosphate ion has a charge of $2-$.

10. Perchlorate ion has more oxygen atoms than chlorate ion.

Section 3: Multiple Choice

11. Which of the following is the correct formula for nitrate?

- a) NO_2^-
- b) NO_3^-
- c) NH_3
- d) HNO_3

12. Which polyatomic ion has a $3-$ charge?

- a) Phosphate
- b) Sulfite

- c) Dichromate
- d) Chlorite

Section 4: Fill in the Blanks

16. The ion HCO_3^- is called ______________.

17. The formula for the sulfite ion is ______________.

18. Write the charge on the cyanide ion: ______________.

Section 5: Short Answer

21. Write the formula for the phosphate ion.

22. Name the ion $C_2H_3O_2^-$.

Section 6: Application

26. Write the dissociation equation for K_2CO_3 in water.

27. Write the balanced chemical equation for the reaction between $NaOH$ and H_2SO_4.

Section 7: Matching Charges

31. Match the following ions to their correct charges:

- a) Phosphate
- b) Hydroxide

Section 8: Advanced Applications

32. Predict whether $CaCl_2$ and Na_2CO_3 will form a precipitate when mixed.

Answer Key

1. a-i, b-ii, c-iii, d-iv, e-v

2. a-i, b-ii, c-iii, d-iv, e-v

3. True

4. True